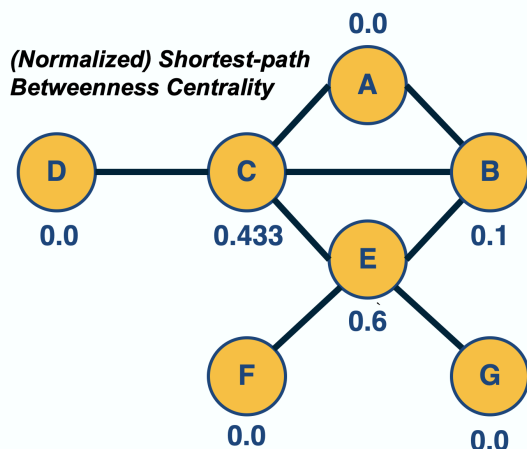


L6: Betweenness Centrality Variants

Shortest-path Betweenness Centrality



In some network analysis applications, the importance of a node is associated with how many paths go through a node: the more routes go through a node, the more central that node is.

The most common instance of this metric is the “shortest-path betweenness centrality”. Consider any two nodes s (source) and t (target) in the same connected component, and let us define that the number of shortest-paths between these two nodes is $n_{s,t}$. Also, suppose that the subset of these paths that traverses node i is $n_{s,t}(i)$, where node i is different than s and t .

The (shortest-path) betweenness centrality of node i is defined as:

$$v_i = \sum_{s,t \neq i} \frac{n_{s,t}(i)}{n_{s,t}}$$

If s and t are the same, we define that $n_{s,t} = 1$.

The previous metric is often normalized by its maximum possible value (which is $(n-1)(n-2)$, for a star network with n nodes) so that the centrality values are between 0 and 1. This is not necessary however given that we only care about the relative magnitude of centralities.

Note that node E, in the visualization, has higher betweenness centrality than node C – even though the two nodes have the same closeness centrality (and node C has higher eigenvector centrality than node E). The reason is that node E is the only “bridge” between nodes F and G and the rest of the network.

For weighted networks, where the (non-negative) weight represents the cost of an edge, the shortest-paths can be computed using Dijkstra’s algorithm for weighted networks.

There are many variants of the betweenness centrality metric, depending on what kind of paths we use. One such variant is the **flow betweenness centrality** in which we compute the max-flow from any source node s to any target node t – and then replace the fraction $\frac{n_{s,t}(i)}{n_{s,t}}$ with the fraction of the max-flow that traverses node i – see Lesson-2 if you do not remember the definition of max-flow.

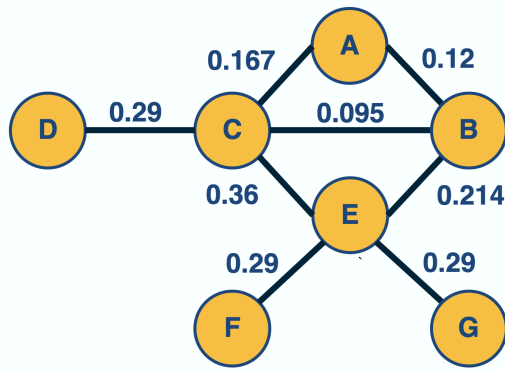
Yet another variant is the **random-walk betweenness centrality**, which considers the number of random walks from node- s to node- t that traverse node- i .

Edge Centrality Metrics

In some cases, we are interested in the centrality of edges, rather than nodes. For example, suppose you are given a communication network and you want to rank links based on how many routes they carry.

One such metric is the edge betweenness centrality. The definition is the same as for node betweenness centrality – but instead of considering the fraction of paths that traverse a node i , we consider the fraction of that traverse an edge (i,j) . These paths can be shortest-paths or any other well-defined set of paths, as we discussed on the previous page.

Another way to define the centrality of an edge is to quantify the impact of its removal. For instance, one could measure the increase in the **Characteristic Path Length** (<https://gatech.instructure.com/courses/552990/pages/l5-diameter-characteristic-path-length-and-network-efficiency>) (CPL, see Lesson-5) after removing edge (i,j) – the higher that CPL increase is, the more important that edge is for the network.



**Shortest-path Edge
Betweenness Centrality**

Food For Thought

An interesting question is how to compute the shortest-path betweenness centrality metric efficiently. We recommend you review the following paper for an efficient algorithm:

Ulrik Brandes: A Faster Algorithm for Betweenness Centrality. Journal of Mathematical Sociology 25(2):163-177, 2001

(<https://doi.org/10.1080/0022250X.2001.9990249> )

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